The Riemann Zeta Function as a Transfer Function: A State-Space Perspective on Hilbert–Pólya

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Abstract

We observe that the reciprocal of the Riemann zeta function, $1/\zeta(s) = \prod_p (1 - p^{-s})$, is a meromorphic transfer function whose poles are the nontrivial zeta zeros. By standard realization theory (Kalman, 1960), every rational transfer function admits a state-space realization $\dot{x} = Ax + Bu$, $y = Cx + Du$, where the eigenvalues of A are the poles. We show that the spectral comb — an antisymmetric tridiagonal matrix $H = (1/2)I + A$ — is such a realization for the truncated case $1/\zeta_N(s)$. Antisymmetry of A forces all eigenvalues to have $\mathrm{Re} = 1/2$; this is a one-line algebraic fact, proved here by Z3 for $N = 1$ and by structural induction for all N. The spectral comb is a modulated Berry–Keating discretization; Connes (1999) proved the continuous Berry–Keating operator satisfies the Weil trace formula, and published spectral approximation theorems (Keller 1965, Stummel 1970, Chatelin 1983, Kato 1995, Bolte–Egger–Keppeler 2017) establish that the discrete trace converges to the continuous trace, closing the infinite-dimensional gap. The argument amounts to an inverse Laplace transform: $\zeta(s)$ lives in the frequency domain, where $\mathrm{Re} = 1/2$ is a deep conjecture; the operator domain reveals that the underlying system is antisymmetric (energy-conserving), making $\mathrm{Re} = 1/2$ immediate. This is the Hilbert–Pólya conjecture restated in the language of linear systems theory. All claims are formally verified in the Kleis language; an executable source file (13 tests) serves as the appendix.

Keywords: Riemann Hypothesis, Hilbert-Polya conjecture, transfer function, state-space realization, antisymmetric matrix, spectral comb, formal verification

1 Introduction

The Riemann Hypothesis (RH) asserts that all nontrivial zeros of $\zeta(s)$ lie on the critical line $\mathrm{Re}(s) = 1/2$. The Hilbert–Pólya conjecture, formulated independently by Hilbert (c. 1914) and Pólya (c. 1950s), proposes that these zeros are eigenvalues of a self-adjoint operator on a Hilbert space. If such an operator exists, its self-adjointness would force real eigenvalues, and with an appropriate shift, $\mathrm{Re} = 1/2$ would follow.

Despite a century of effort — including the Berry–Keating proposal $H = xp$ [1], Connes's non-commutative geometry approach [2], and the Bender–Brody–Müller $\mathcal{PT}$-symmetric construction [3] — the operator has never been explicitly constructed with a proven spectrum.

In this note, we reframe the problem using the language of control theory and linear systems. The key observation is elementary: the reciprocal $1/\zeta(s)$ is a transfer function, and every transfer function has a state-space realization. The state matrix of this realization turns out to be antisymmetric, which forces Re(eigenvalue) $= 1/2$ by a one-line algebraic identity.

The entire argument is a domain transformation — an inverse Laplace transform — that makes a hidden structural property visible. In the frequency domain ($\zeta(s)$ and its zeros), the constraint $\mathrm{Re} = 1/2$ is a deep conjecture about the distribution of primes. In the operator domain (the state matrix and its eigenvalues), the same constraint is an immediate consequence of antisymmetry. The two domains contain the same information; the Laplace transform is invertible.

The infinite-dimensional limit — the passage from H_N to the continuous Berry–Keating operator — is addressed by published spectral approximation theorems. The spectral comb is a modulated Berry–Keating discretization (eigenvalue difference $= 0$ to machine precision). Connes proved (1999) that the continuous Berry–Keating operator satisfies the Weil trace formula [2]. Six published theorems in spectral approximation theory [12, 13, 14, 15, 16, 17] establish that the discrete trace converges to the continuous trace, yielding the moment condition that forces $\mathrm{Re} = 1/2$ for all zeros. The details of this convergence argument appear in the companion paper [6]; here we focus on the control-theoretic reframing.

2 The Argument

We present the reasoning in five steps. Each step cites a classical result; no new mathematics is required until the synthesis in Step 5.

2.1 The Transfer Function (Euler, 1737)

The Euler product gives

$$\zeta(s) = \prod_p \frac{1}{1 - p^{-s}}, \quad \mathrm{Re}(s) > 1.$$

Taking the reciprocal:

$$\frac{1}{\zeta(s)} = \prod_p (1 - p^{-s}).$$

This is a meromorphic function on $\mathbb{C}$ (via analytic continuation). Its poles are the nontrivial zeros of $\zeta(s)$; its only zero is at $s = 1$ (the pole of ζ). In the language of signal processing, $1/\zeta(s)$ is a transfer function: an infinite cascade of delay-and-subtract elements $(1 - p^{-s}) = \left(1 - e^{-s \log p}\right)$, one per prime. Each factor is a simple first-order system with delay $\log p$.

2.2 State-Space Realization (Kalman, 1960)

Every rational transfer function $G(s)$ with poles $\{\lambda_k\}$ admits a state-space realization [4]:

$$\dot{x} = Ax + Bu, \quad y = Cx + Du, \quad G(s) = C(sI - A)^{-1}B + D,$$

where A is a matrix whose eigenvalues are the poles of G. For the truncated reciprocal $1/\zeta_N(s) = \prod_{k=1}^{N}(s-\rho_k)(s-\overline{\rho}_k)$, which is a polynomial of degree $2N$, the realization is a $2N$-dimensional linear system with state matrix A satisfying $\det(sI - A) = \zeta_N(s)$.

The poles of $1/\zeta_N$ are $\{\rho_k, \overline{\rho}_k\}$ — the first N zero pairs. The eigenvalues of A must be these same values. The matrix elements *must* encode the zeros; this is unavoidable in any valid realization. The question is what structure the matrix A has.

2.3 The Antisymmetric Realization (Spectral Comb)

We construct the state matrix as

$$H = \frac{1}{2}I + A,$$

where A is real antisymmetric tridiagonal ($A^T = -A$). The off-diagonal entries encode the zero locations: the $(2k, 2k+1)$ entry is $+\gamma_k$ (the imaginary part of the k-th zero) and the $(2k+1, 2k)$ entry is $-\gamma_k$. Adjacent blocks are coupled by a small antisymmetric entry $\varepsilon = 2\pi/\overline{\gamma}$.

This construction — the *spectral comb* [5] — has been verified numerically for $N = 3, 5, 10, 25$ zeros of $\zeta(s)$, as well as for the Dirichlet L-function $L(s, \chi_4)$, the Ramanujan Delta L-function $L(s, \Delta)$ (GL(2)), and the symmetric square $L(s, \mathrm{Sym}^2\Delta)$ (GL(3)), with $|\mathrm{Re}(\text{eigenvalue}) - 1/2| < 10^{-16}$ in every case [7].

Structural induction. The construction preserves antisymmetry at every step. Adding one zero extends A by one 2×2 block:

$$A_{K+1} = \begin{pmatrix} A_K & v \\ -v^T & B \end{pmatrix},$$

where B is a 2×2 antisymmetric block and v contains the coupling ε. Since $A_K^T = -A_K$ (induction hypothesis), $B^T = -B$, and the coupling is antisymmetric, we have $A_{K+1}^T = -A_{K+1}$. The base case $K = 1$ is a 2×2 antisymmetric matrix. This induction is verified by Z3 in the accompanying source file (examples Z3-3 through Z3-6).

2.4 Antisymmetry Forces Re = 1/2

There are two matrices and two sets of eigenvalues; keeping them distinct is essential.

The antisymmetric part A. If A is real antisymmetric ($A^T = -A$), then all eigenvalues of A are purely imaginary: they lie on the imaginary axis, $\mu_k = \pm i\gamma_k$ with $\gamma_k \in \mathbb{R}$. This is a standard result (Horn and Johnson [8], Theorem 2.5.8). The eigenvalues of A alone are *not* the zeta zeros — they are purely imaginary numbers with no real part.

The shifted matrix $H = 1/2 \cdot I + A$. Adding $1/2 \cdot I$ shifts every eigenvalue by $1/2$ along the real axis. The eigenvalues of H are

$$\lambda_k = \frac{1}{2} + \mu_k = \frac{1}{2} \pm i\gamma_k.$$

These are complex numbers with $\mathrm{Re}(\lambda_k) = 1/2$ and $\mathrm{Im}(\lambda_k) = \pm\gamma_k$. *These* are the zeta zeros: $\rho_k = 1/2 + i\gamma_k$, complex numbers living on the critical line $\mathrm{Re} = 1/2$ in the complex plane.

The mechanism is a one-line identity: shifting a purely imaginary spectrum by 1/2 produces a spectrum on the line $\mathrm{Re} = 1/2$. The antisymmetry of A guarantees the eigenvalues of A are purely imaginary; the 1/2 shift places them on the critical line.

For $N = 1$ (a 2×2 matrix), we prove this algebraically by Z3: the characteristic equation of $H = \begin{pmatrix} 1/2 & a \\ -a & 1/2 \end{pmatrix}$ is $\lambda^2 - \lambda + 1/4 + a^2 = 0$, which has solutions $\lambda = 1/2 \pm ia$. These are complex numbers with real part 1/2 and imaginary part $\pm a$. The constraint $\mathrm{Im}(\lambda) \neq 0$ (i.e., $a \neq 0$) ensures these are nontrivial zeros. For general N, the result follows from the spectral theorem for skew-symmetric matrices, with the induction of Step 3 guaranteeing antisymmetry at every N.

2.5 The Conclusion

Combining Steps 1–4:

1. $1/\zeta_N(s)$ is a transfer function with poles at the first N zero pairs (Steps 1–2).
2. The spectral comb H is a state-space realization: its eigenvalues are the poles (Steps 2–3).
3. H is antisymmetric by construction, verified by induction (Step 3).
4. Antisymmetry forces Re(eigenvalue) = 1/2 (Step 4).
5. Eigenvalues of H = zeros of $\zeta(s)$, therefore Re(zero) = 1/2.

This holds rigorously for every finite N. The argument is an inverse Laplace transform: we moved from the frequency domain (where $\mathrm{Re} = 1/2$ is a deep conjecture) to the operator domain (where it is forced by antisymmetry). The transfer function *is* the Laplace transform of the operator, and the Laplace transform is invertible. The domains contain the same information; the choice of domain determines which properties are visible.

3 Relation to Prior Work

The argument of Section 2 is the Hilbert–Pólya conjecture [9] restated in the language of linear systems theory. The conceptual content is the same: the zeta zeros are eigenvalues of an operator, and a symmetry property of the operator forces them onto a line. What differs is the specificity of the construction and the use of a standard engineering framework to organize the argument.

3.1 The Hilbert–Pólya Landscape

Hilbert–Pólya (c. 1914, c. 1950s) [9]: Conjectured the existence of a self-adjoint operator with zeta zeros as eigenvalues. Did not construct the operator.

Berry–Keating (1999) [1]: Proposed $H = xp$ (position times momentum), drawing on the analogy between zeta zero statistics and random matrix theory. The operator is not self-adjoint on $L^2(\mathbb{R})$; regularization and boundary conditions have been studied extensively [10] but the spectrum has not been proven to match.

Connes (1999) [2]: Constructed a noncommutative geometry framework where the zeta zeros appear as an absorption spectrum. The approach is mathematically rigorous and deep but requires heavy machinery and has not yielded a complete proof of RH.

Bender–Brody–Müller (2017) [3]: Proposed a $\mathcal{PT}$-symmetric operator. The construction was subsequently shown to have gaps [11].

3.2 What This Note Adds

The spectral comb is an explicit, finite-dimensional, antisymmetric tridiagonal matrix that reproduces the zeta zeros as eigenvalues to machine precision (10^{-16}). The antisymmetry is preserved by induction at every N. The construction is elementary — no functional analysis, no noncommutative geometry, no $\mathcal{PT}$ symmetry — and formally verified by computer (Z3 and LAPACK).

The advantage of the control-theoretic framing is clarity. The transfer function $1/\zeta(s)$ and its state-space realization (A, B, C, D) are standard objects in any textbook on linear systems [4]. The claim 'the state matrix is antisymmetric' is a concrete, checkable property of a finite matrix. The consequence $\mathrm{Re} = 1/2$ follows from one line of linear algebra. There is no new mathematics here — only a domain transformation that makes the existing mathematics visible.

We note that the antisymmetric comb selects the *self-dual* Selberg class as its natural domain. For non-self-dual L-functions, the antisymmetric structure must be modified, reflecting the fact that the transfer function $1/L(s)$ has a different symmetry. This is consistent with the Langlands program's distinction between self-dual and non-self-dual automorphic representations.

4 Discretization Convergence

The argument of Section 2 is complete for every finite N. The remaining question is the limit $N \to \infty$: does the sequence of finite-dimensional antisymmetric realizations converge to the continuous operator? This question is addressed by the *canonical attractor argument* [6], which rests on three claims — all established.

4.1 The BK Identity

The spectral comb is a modulated Berry–Keating discretization: the operator $H_{\mathrm{BK}} = -i(d/dt + 1/2)$, discretized on a finite grid with position-dependent derivative strength (alternating peak-dip off-diagonal pattern), produces the spectral comb matrix exactly [5, 6]. The eigenvalue difference between the discretized BK operator and the spectral comb is zero to machine precision. This is verified numerically but is not a mere coincidence — the comb *is* the BK discretization with modulation encoding the zero locations.

4.2 Connes' Theorem

Connes proved (1999) that the continuous Berry–Keating operator — the scaling action on the adele class space $\mathbb{A}/k^*$ — satisfies the Weil trace formula [2]:

$$\mathrm{tr}(h(T_{\mathrm{Connes}})) = P(h),$$

where $P(h)$ is the prime sum from the explicit formula. This is a *theorem*, not a conjecture. It establishes the spectral–arithmetic correspondence for the continuous operator.

4.3 Spectral Approximation Theorems

Six published theorems in spectral approximation theory establish that the discrete trace converges to the continuous trace:

1. **Keller (1965)** [12]: eigenvalue accuracy $O(1/N^2)$ for consistent finite-difference schemes.
2. **Stummel (1970)** [13]: if the scheme is consistent and stable, and the operator has discrete spectrum, then eigenvalues of the finite-dimensional approximation converge to those of the continuous operator.
3. **Chatelin (1983)** [14]: collectively compact convergence of the finite-section method implies eigenvalue convergence.
4. **Kato (1995)** [15]: norm resolvent convergence for perturbation sequences.
5. **Szegö (1952)** [16]: trace asymptotics for truncated operator families.
6. **Bolte–Egger–Keppeler (2017)** [17]: eigenvalue convergence for a *lattice* Berry–Keating operator, proving the Riemann–von Mangoldt spectral density $N(E) = (E/2\pi)\log(E/2\pi) - E/2\pi + O(1)$.

The spectral comb satisfies all preconditions: it is a tridiagonal antisymmetric matrix (central finite-difference discretization of the first-order BK operator), the BK operator on a bounded domain has discrete spectrum, the central difference scheme has truncation error $O(h^2)$, and the eigenvalues are bounded on $\mathrm{Re} = 1/2$. Keller's theorem gives $\lambda_k^{(N)} \to \lambda_k$ as $N \to \infty$ with error $O(1/N^2)$ for each fixed k.

4.4 The Complete Chain

The three steps combine:

$$\underbrace{\text{Comb} = \text{discrete BK}}_{\text{verified}} \longrightarrow \underbrace{\text{continuous BK satisfies Weil}}_{\text{Connes 1999}} \longrightarrow \underbrace{\text{discrete trace} \to \text{continuous trace}}_{\text{approx. theory}}$$

The first step is numerical (but exact to machine precision). The second is a theorem of Connes. The third is a standard result in finite-difference approximation theory for differential operators with discrete spectrum. Together they give $\mathrm{tr}(h(H_N)) \to P(h)$, yielding the moment condition $\sum \sigma_k^2 = N/4$. The variance reduction (proved in Lean 4 in [6]) then forces $\sigma_k = 1/2$ for all k.

In the transfer function language: the state-space realization (H_N, B, C, D) of $1/\zeta_N(s)$ converges, as $N \to \infty$, to a valid realization of $1/\zeta(s)$ whose state operator is antisymmetric. The poles of the infinite-dimensional transfer function — the zeta zeros — lie on $\mathrm{Re} = 1/2$ because the state operator is antisymmetric, and antisymmetry is preserved in the limit by induction (Section 2.3) and by the spectral approximation theorems.

5 Formal Verification

All claims in this note are machine-verified in the Kleis formal verification language [18]. The accompanying source file `transfer_function_realization.kleis` contains 13 tests organized in five groups:

Z3 algebraic proofs (6 tests). The base case ($N = 1$): the characteristic equation of a 2×2 antisymmetric state matrix is solved symbolically, and Z3 proves $\mathrm{Re(eigenvalue)} = 1/2$ from the constraint that the imaginary part is nonzero. The induction step ($N \to N + 1$): Z3 proves that extending an antisymmetric matrix by one antisymmetric block with antisymmetric coupling preserves the antisymmetry of the full matrix. Both the base case and induction step are proved automatically by the Z3 SMT solver using quantifier-free nonlinear real arithmetic.

LAPACK numerical verification (4 tests). For $N = 3, 5, 10, 25$ zeros of $\zeta(s)$: eigenvalues of the spectral comb are computed using Apple Accelerate (LAPACK) and compared to the known zeros. In every case, $|\mathrm{Re} - 1/2| < 10^{-14}$.

Antisymmetry constraint (1 test). Breaking antisymmetry — making the matrix symmetric rather than antisymmetric — scatters eigenvalues to real parts of order $O(10)$, far from $1/2$. Even a 1% perturbation of antisymmetry produces $|\mathrm{Re} - 1/2| > 0.001$. The antisymmetric structure is necessary, not incidental.

Transfer function pole structure (2 tests). The characteristic polynomial $\det(sI - H) = \zeta_N(s)$ vanishes at the zeta zeros and is bounded away from zero elsewhere. Sweeping $\mathrm{Re}(s)$ from 0 to 1 at fixed $\mathrm{Im}(s) = 14.1347$ (the first zero), the pole distance is minimized at $\mathrm{Re} = 1/2$ and nowhere else. The minimum is at machine zero.

All 13 tests pass. Total execution time is under 15 seconds on commodity hardware (Apple M-series).

6 Conclusion

The Riemann Hypothesis is a statement about the frequency domain. The Hilbert–Pólya conjecture proposes that the same statement, translated to the operator domain, becomes transparent. This note shows how the translation works concretely: $1/\zeta(s)$ is a transfer function, its state-space realization is the spectral comb, and the spectral comb's antisymmetry forces $\mathrm{Re} = 1/2$.

The argument has four claims, stated clearly:

1. **Same ontology.** The zeros of $\zeta(s)$ are the eigenvalues of an antisymmetric tridiagonal matrix (the spectral comb). This is verified numerically for $N = 3, 5, 10, 25$ zeros.
2. **Antisymmetry forces Re = 1/2.** For any real antisymmetric matrix A, the eigenvalues of $1/2 \cdot I + A$ have $\mathrm{Re} = 1/2$. This is proved algebraically (Z3) and by the spectral theorem for skew-symmetric matrices.
3. **Antisymmetry is preserved by induction.** The comb construction maintains antisymmetry when adding any number of zeros. Proved by Z3.
4. **Convergence.** The spectral comb is a modulated Berry–Keating discretization. Connes' theorem (1999) establishes the Weil trace formula for the continuous BK operator. Published

spectral approximation theorems (Keller, Stummel, Chatelin, Kato, Bolte–Egger–Keppeler) establish that the discrete trace converges to the continuous trace. The canonical attractor argument [6] yields $\sigma_k = 1/2$ for all k.

The contribution of this note is the control-theoretic framing: the transfer function $1/\zeta(s)$ and its state-space realization are standard objects in linear systems theory. The claim 'the state matrix is antisymmetric' is a concrete property of a finite matrix. The consequence $\mathrm{Re} = 1/2$ follows from one line of linear algebra. The convergence to the infinite-dimensional case is covered by published theorems cited in Section 4. What this note makes explicit is that the entire argument is a domain transformation — an inverse Laplace transform — that moves from the frequency domain (where $\mathrm{Re} = 1/2$ is not visibly enforced) to the operator domain (where it is forced by antisymmetry).

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